

Encryption/Decryption Using XOR

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Objective

This activity was designed to develop an understanding of how XOR encryption functions, by encrypting and decrypting a message with a secret key.

You need to select a Secret Message and Key.

Message: code

Key: safe

There are the same number of letters in both words.

Next, encode the Message in ASCII Binary. Now encode the Message in ASCII Binary.

Character	ASCII Decimal	Binary
c	99	01100011
o	111	01101111
d	100	01100100
e	101	01100101

Finally, convert the Key to ASCII Binary. Now convert Key to ASCII Binary (Step 3).

Character	ASCII <100	ASCII Decimal	ASCII Binary
s	115	01110011	
a	97	01100001	
f	102	01100110	
e	101	01100101	

Next, you need to perform XOR Encryption. After that, you need to do XOR Encryption.

c XOR s

01100011
XOR 01110011
= 00010000

o XOR a

01101111
XOR 01100001
= 00001110

d XOR f

01100100
XOR 01100110
= 00000010

e XOR e

01100101
XOR 01100101
= 00000000

Write the encrypted binary into hexadecimal. Write encrypted binary to hex.

Binary	Hexadecimal
00010000	10
00001110	0E
00000010	02
00000000	00

Encrypted Message:

10 0E 02 00

Decryption Process

The encrypted binary values were then again XORed with the same key to decrypt the message.

Example

```
00010000
XOR 01110011
= 01100011 = c
```

If this is repeated for every character, the following are obtained:

code

The deciphered message is the same as the original plain text message.

XOR with Different Lengths

If the plaintext and the key are not same length the key can be repeated until it is. This enables the XOR operation to proceed until the entire plaintext is done.

Flowchart

```
graph TD
    Start([Start]) --> Choose[Choose message and key]
    Choose --> Translate1[Translates a message to binary.]
    Translate1 --> Translate2[Translates a message to binary.]
    Translate2 --> Convert[The key is converted to binary.]
    Convert --> XOR[Apply XOR operation]
    XOR --> Hex[Convert result to Hex.]
    Hex --> Decrypt[Use XOR and key to decrypt]
    Decrypt --> Verify[Verify original message]
    Verify --> End([End])
```

Challenges Encountered

During this activity one issue that had to be really careful about was the application of the XOR operation, as it was simple to confuse the binary digits. The other challenge was reading the binary values with encryption and changing them to hexadecimal. Each step was carefully examined and I was able to carry out the encryption and then decryption correctly. The activity was a good way to gain insight into the operation of XOR encryption in cybersecurity.